Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/24/2006
PI Name:	Hegseth, John J. Ph.D.		
Project Title:	Geophysical Flow Experiments in Rotating Spherical Capacitors		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	70148	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	10/01/1999	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	note: extended 1 year		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2447		
Performance Goal No.:			
Performance Goal Text:			

We are studying the flow patterns and instabilities that are seen in ground based experiments when a temperature difference and a voltage difference is applied between two spheres. To make the system geophysically similar (similar to a planetary atmosphere, an ocean, a mantle, an outer core, etc.), a Coriolis force will then be applied to the system by performing the experiments on a rotating table. Shadowgraph flow visualization is used. Shadowgraph visualization will also be tested in a near-critical fluid convection system using a rotating table. This highly compressible fluid will produce an even more realistic geophysical analog in weightlessness. Other experiments will also be performed using the near-critical fluid cell.
We were going to do the following three things. 1) Provide the information necessary to build a Rotating Spherical Capacitor Apparatus (RoSCA). 2) Test the transmission and reflection shadowgraph technique in the rotating frame of a rotating laboratory table. 3) Prepare and possibly present a successful Science Concept Review so that we can start the

	process of building a flight model RoSCA.
Task Description:	The work intends to provide a laboratory apparatus for experiments that enables the study of planetary geophysical flows. This laboratory apparatus will greatly improve the state of the art by observing flow phenomena in the rotating frame with a compressible fluid media. Having an experimental model of the atmosphere, oceans, and liquid cores to complement the many computer models is expected to improve understanding of many global flow phenomena (e.g., weather patterns, core convection, ocean circulation). Previous experiments have studied oscillatory cellular structures in the incompressible fluid. These structures will be analogous to the structures seen in large-scale planetary flows when a Coriolis force is applied. Previous experiments also showed that in a space experiment in a compressible near-critical fluid we could study both compressibility effects and latent heat effects. Both of these effects are very important in geophysical flows such as planetary atmospheres and planetary liquid cores. Furthermore we know of no other system that is capable of studying these effects. These previous results could not produce phenomena of geophysical significance because they were not performed in a rotating frame of reference. Similar experiments with the fictious forces applied to the system in a rotating frame will be geophysically similar. This project will bring this system into the rotating frame. This will provide the necessary information for the similar microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)